AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-AOH-I035*

Team Name: *Spark Titans*

Domain: *Food security,Agriculture and rural Transformation*

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Presentation Outline

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Problem Statement

Deduction of crop diseases at early stage using AI - model and hyperspectral camera at affordable technology.

Problem Understanding

- Small farmers who often face difficulties in identifying crop diseases early and cannot afford expensive detection technologies are the main target users of our solution. They need a simple, affordable, and reliable way to check the health of their crops.
- The general causes for this problem is Limited expert support, high technology costs, low awareness, and similar disease symptoms.
- Farmers can detect diseases early, protect their crops, and reduce losses. This helps the users or stakeholders to save money, maintain a stable income, and use pesticides more wisely while ensuring a better food supply for everyone.

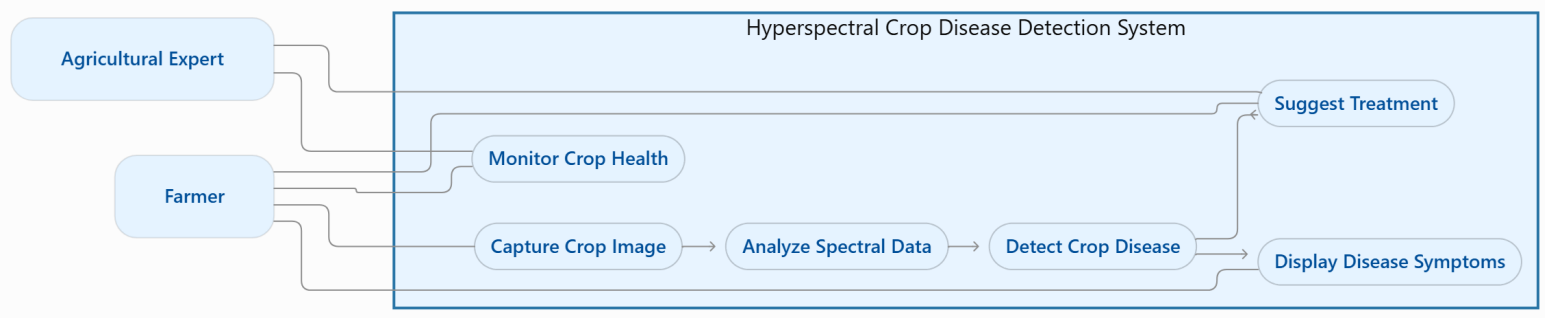
Existing Gaps & Challenges

- Farmers often find it difficult to identify crop diseases at an early stage because the symptoms are not always clearly visible to the naked eye. By the time they notice changes such as yellow leaves, dark spots, or unusual plant growth, the disease may have already spread and damaged a large part of the crop. This can lead to reduced crop production, financial losses, and difficulties in maintaining healthy plants. Therefore, detecting diseases early is very important to help farmers protect their crops and improve their yield..
- Similar diseases can be difficult to distinguish, while poor internet connectivity and limited digital skills create additional challenges.
- Agricultural experts may not be easily accessible in every village, and detection tools can be costly or complicated to use.

Our Proposed Solution

- Our project focuses on helping farmers identify crop diseases before they cause serious damage. It uses a hyperspectral camera to capture details about plants that cannot be easily seen with the naked eye. .
- It works by analysing the light reflected from the crops, the system can identify early signs of disease or plant stress. This allows farmers to understand their crop health and take action at the right time. The main aim is to make disease detection easier and more effective, reduce crop losses, save money, and help farmers grow healthier crops using smarter and more sustainable methods
- Our project helps farmers detect crop diseases early using a hyperspectral camera. It helps reduce crop losses, save money on pesticides, and improve crop health. The main goal is to make farming easier, smarter, and more sustainable.

Technical Approach



The hyperspectral camera collects the data, the frontend Streamlit accepts it, Python processes it, the AI model predicts the disease, SQLite saves the results, and the frontend shows those results to the farmer.

Tech Stack

Frontend: Streamlit

Backend: Python

Database: SQLite

AI/ML, APIs, or Hardware: N/A

Innovation & Uniqueness

- Our method uses a hyperspectral camera to detect crop diseases early, even before visible symptoms appear. Unlike normal cameras, it captures different wavelengths of light to identify plant health problems. This helps farmers save time, reduce crop losses, avoid unnecessary pesticide use, and improve crop health.

- This approach is suitable because it helps detect crop diseases early, even before visible symptoms appear. It helps farmers take quick action, reduce crop losses, save money, and use pesticides only when needed. It makes farming easier and more efficient.

- Our main difference is that we use a hyperspectral camera to identify crop diseases before clear symptoms appear. It helps farmers spot problems early, save time, reduce crop losses, and take better care of their crops.

Feasibility & Scalability

- **Feasibility:** We need a hyperspectral camera, a computer, Python, an AI model, and an SQLite database. First, we collect crop images, then process the data and train the model to detect diseases. Finally, we connect everything to a simple website that shows the detected disease and its symptoms.

- **Scalability:** Our project can be improved by detecting more crop diseases and supporting different types of crops. In the future, we can develop a mobile app, cover larger farms, and provide farmers with faster and more accurate results.

- **Risks:** The main challenges are the high cost of hyperspectral cameras, limited training data, and the need for accurate results. We can overcome these by using affordable equipment, collecting more crop images, and improving the AI model through regular testing.

Cost for Product (Optional)

- Development or hardware cost: 5,000–15,000Rs
- Hosting or maintenance cost: 0–2,000Rs
- Estimated total: 5,000–17,000Rs

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Uploads crop images captured using a hyperspectral camera to detect diseases early, understand symptoms, and take timely action to protect crops.
- Use case 2: Analyses the disease detection results to understand plant health problems and recommend suitable treatments to farmers.
- Use case 3: Monitors the health of crops regularly, keeps track of disease detection results, and takes preventive measures to reduce crop losses and improve productivity.

Impacts and Benefits

- Our project helps farmers identify crop diseases at an early stage, save time, reduce crop losses, and take better care of their crops.
- It can help farmers save money, reduce the unnecessary use of pesticides, protect the environment, and support healthier and more sustainable farming.
- We aim to detect diseases faster, reduce crop damage and pesticide use, and improve crop productivity through early detection and timely treatment.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.4	Zero Hunger	Our project helps farmers detect crop diseases early, reduce crop losses, and improve food production.
SDG 12	Target 12.4	Responsible Consumption and Production	It helps reduce unnecessary pesticide use and encourages safer, more sustainable farming practices.

Individual Member Contribution

Member	Contribution or responsibility
Raja Raja Shree.A	Team Leader
Rithvik.K	Research and information gathering
Rubiga.R.K	Coding and research
Deepak E R	Power point presentation

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